# Retrieval of Land Surface Temperature, Emissivity, and Atmospheric Parameters From Hyperspectral Thermal Infrared Image Using a Feature-Band Linear-Format Hybrid Algorithm

Huazhong Ren, Xin Ye, Jing Nie, Jinjie Meng, Wenjie Fan, Qiming Qin, Yanzhen Liang, and Hongcheng Liu

***Abstract*— Thermal infrared remote sensing can acquire large-scale land surface thermal radiance effectively. However, the observed data are affected by surface and atmospheric conditions. Traditional methods require some prior knowledge, such as emissivity in split-window algorithm and atmospheric correction in temperature–emissivity separation algorithm. This information is difficult to obtain directly and accurately. Hyperspectral thermal infrared data provide the possibility for simultaneous retrieval of atmospheric parameters, land surface temperature (LST), and emissivity because of their abundant band information. This study proposed a feature-band linear-format hybrid (FebLihy) algorithm by combining a deep neural network (DNN) model and a physical model with thermal airborne hyperspectral imager (TASI) data. The proposed algorithm was divided into three steps. First, the radiative transfer equation was converted into a linear form, and seven feature bands were chosen to reduce the unknowns. Second, the initial values of atmospheric and land surface parameters were estimated with the DNN model. Finally, least-squares optimization was used in the physical model to retrieve the final results. Results of the simulation data showed that the root-mean-square error (RMSE) of LST was 0.86 K, the RMSE of emissivity was less than 0.015, and the accuracy of atmospheric parameters was improved effectively by the physical model. The FebLihy algorithm was applied in a real TASI image in Fuyun County and verified with CE312 ground measurement data. Accurate results were achieved. The FebLihy algorithm will be optimized in terms of model and data in the future study.**

***Index Terms*— Atmospheric thermal radiance, atmospheric transmittance, land surface emissivity, land surface temperature (LST), thermal infrared hyperspectral.**

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## I. Introduction

LAND surface temperature (LST) is a key parameter used in the physics of land surface processes regionally and globally. It has been utilized in soil moisture estimation and climatic, hydrological, ecological, and biogeochemical studies. Thermal infrared remote sensing provides a promising way to obtain LST at large spatial and temporal scales and has attracted much attention in the last four decades. To date, many LST retrieval methods have been proposed in accordance with thermal infrared radiometer (TIR) instruments [1]–[10]. These methods include the single-band algorithm, multiband algorithms (e.g., split-window algorithm and temperature–emissivity separation algorithm), and multitime algorithms (e.g., temperature-independent spectral index algorithm [11] and physical day and night algorithm [12]). The coupling of land surface and atmosphere in thermal radiation makes the accurate retrieval of LST need atmospheric data as *a priori* knowledge to remove the atmospheric effect and vice versa. However, the atmospheric data cannot be obtained directly for most TIR sensors. They depend on other resources, such as the reanalysis atmospheric database of National Centers for Environmental Prediction (NCEP) [13] and European Center for Medium-Range Weather Forecasts (ECMWF) [14] or atmospheric profiles from satellite observation. The discrepancy between the observation of TIR sensors with the atmospheric data in time and space causes uncertainty for the LST retrieval. Apart from the atmospheric effect, the retrieval of LST has to remove the emissivity effect, and many previous studies have investigated this topic [6], [15]–[18].

In the recent decade, hyperspectral thermal infrared sensors, such as Atmospheric Infrared Sounder (AIRS) [19] and Infrared Atmospheric Sounding Interferometer (IASI) [20], provide thousands of narrow bandwidth bands in thermal wavelengths. These sensors supply much information to simultaneously separate atmospheric data, LST, and emissivity.

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The key step is to reduce the number of unknowns to make it equal to or less than that of observation equations. A few studies based on empirical models or physical models have been conducted in recent years [21]–[28]. The physical model is time-consuming because it always has to derive the atmospheric radiative transfer model to calculate the atmospheric radiance in TIR and can provide highly accurate results. The empirical model runs faster but with a relatively lower accurate result. The AIRS and IASI are nonimaging sensors, and the current retrieval methods mainly deal with the point data rather than the image. Their efficiency decreases when applied to image data.

The thermal airborne hyperspectral imager (TASI) [29], manufactured by the ITRES company in Calgary, AB, Canada, is an airborne push-broom thermal imager with 32 bands in 8.0–11.5 $\mu$m and features 600 spatial pixels in a field of view (FOV) as 40°. This imager is used to detect LST, emissivity, and conduct applications, such as mineral mapping, land mining, and gas exploration [30]–[33]. Some studies [34]–[36] have used temperature and emissivity separation (TES) algorithm to obtain LST and emissivity from the surface-leaving radiance data of airborne TASI image after atmospheric correction. The used atmospheric profile data can be obtained from the simultaneous radiosonde balloon, which is expensive and is unavailable for each TASI flight, or from the above reanalysis atmospheric profile [37], [38], which cannot match the TASI flight exactly in time and space. Therefore, developing a new algorithm to retrieve LST and emissivity effectively and accurately is required. Inspired by the algorithms used in AIRS and IASI [39]–[44], this study focuses on the simultaneous retrieval of atmospheric parameters (e.g., transmittance, thermal radiance), LST, and emissivity from TASI multiband TIR images by combining deep neural network (DNN) and physical models. This process reduces the number of unknowns in terms of feature bands and increases the computation speed by retrieving atmospheric upward, downward radiance, and transmittance rather than atmospheric vertical profiles. The results are validated with ground-measured data. The rest of this article is organized as follows. Section II describes the proposed algorithm for the retrieval of atmospheric parameters, LST, and emissivity. Section III shows the data set used in this study. Section IV presents the retrieval result and validation through *in situ* observation. Section V gives the discussions. Section VI summarizes the conclusions.

## II. Methodology and Analysis

### A. Radiative Transfer Equation in TIR

For a cloud-free atmosphere, the band-effective radiance observed at the observation height from the surface target can be expressed as

$$L_i = \left[\varepsilon_i B_i(T_s) + (1-\varepsilon_i)L_{ai}^{\downarrow}\right] \cdot \tau_i + L_{ai}^{\uparrow} \tag{1}$$

where $L_i$ is the radiance in the $i$th band, $B_i(T_s)$ is the blackbody radiance following Planck's law at the surface temperature $T_s$, and $\varepsilon_i$ is the band emissivity. $L_{ai}^{\downarrow}$ is the atmospheric downward thermal radiance from space to the ground; $L_{ai}^{\uparrow}$ and $\tau_i$ are the atmospheric upward thermal radiance and transmittance from the ground to the flight height. Equation (1) shows that the retrieval of $T_s$ from $L_i$ has to deal with $\varepsilon_i$, $L_{ai}^{\downarrow}$, $L_{ai}^{\uparrow}$, and $\tau_i$, especially for single-band and multiband TIR sensors. However, for the hyperspectral TIR sensor, $T_s$, $\varepsilon_i$, $L_{ai}^{\downarrow}$, $L_{ai}^{\uparrow}$, and $\tau_i$ can be obtained together. Section II-B–II-D will show a new solution for such simultaneous retrieval.

TABLE I
Center Wavelengths for the 32 Bands of TASI

| No. | $\lambda_c$(μm) | No. | $\lambda_c$(μm) | No. | $\lambda_c$(μm) | No. | $\lambda_c$(μm) |
|---|---|---|---|---|---|---|---|
| 1 | 8.055 | 9 | 8.931 | 17 | 9.807 | 25 | 10.683 |
| 2 | 8.164 | 10 | 9.040 | 18 | 9.916 | 26 | 10.792 |
| 3 | 8.274 | 11 | 9.150 | 19 | 10.026 | 27 | 10.902 |
| 4 | 8.383 | 12 | 9.259 | 20 | 10.135 | 28 | 11.011 |
| 5 | 8.493 | 13 | 9.369 | 21 | 10.245 | 29 | 11.121 |
| 6 | 8.602 | 14 | 9.478 | 22 | 10.354 | 30 | 11.230 |
| 7 | 8.712 | 15 | 9.588 | 23 | 10.464 | 31 | 11.340 |
| 8 | 8.821 | 16 | 9.697 | 24 | 10.573 | 32 | 11.449 |

### B. Feature-Band Empirical Model Development

From (1), one equation has five unknowns ($T_s$, $\varepsilon_i$, $L_{ai}^{\downarrow}$, $L_{ai}^{\uparrow}$, and $\tau_i$). $N$ bands' equations have $4N+1$ unknowns because $T_s$ is considered the same for every band, and then the 32 bands (Table I) of the TASI image correspond to 129 unknowns. The way of reducing the number of unknowns in this article is to establish empirical relationships from the several feature bands to the others, which can be expressed as

$$x_i = \sum_{m=1}^{M} a_{m,i} x_m + b_i \tag{2}$$

where $a_{m,i}$ and $b_i$ are band-dependent coefficients and constant, respectively. $x_i$ is the band value and related to $\varepsilon_i$, $L_{ai}^{\downarrow}$, $L_{ai}^{\uparrow}$, and $\tau_i$. $i$ is the specified band whose value is calculated using that of the $M$ feature bands. Since the physical model is obtained based on the radiative transfer equation, the empirical relationship that preserves the original data was chosen in our study. Besides, the effective value range of each parameter of the $i$th band is limited to ensure validity. For example, the emissivity range is [0.65, 1.00], and atmospheric transmittance range is set as [0.0, 1.0].

As stated above, $M$ feature bands denote $4M+1$ unknowns. The condition that the number of the total band must be larger than the unknowns (i.e., $N \geq 4M+1$) makes $M \leq (N-1)/4$. Given that $N = 32$ for TASI, then $M \leq 7$, indicating that the values ($\varepsilon_i$, $L_{ai}^{\downarrow}$, $L_{ai}^{\uparrow}$, and $\tau_i$) of seven bands at most can be retrieved using (2) from TASI. Thus, only 29 unknowns are left, including one $T_s$ and $\varepsilon_i$, $L_{ai}^{\downarrow}$, $L_{ai}^{\uparrow}$, and $\tau_i$ of the seven feature bands. According to some previous studies [21], [45], the band selection should be implemented to minimize information loss and prevent the effect of errors or uncertainties on retrieval accuracy. To check the above simplification, we performed principal component analysis on $\varepsilon_i$, $L_{ai}^{\downarrow}$, $L_{ai}^{\uparrow}$, and $\tau_i$, and found that the first seven principal components can explain more than 99.9% of the total information.

Fig. 1 shows the spectral atmospheric transmittance of the mid-latitude summer atmospheric model simulated from the MODTRAN5.2 code and the locations of TASI 32 bands. Bands 1–3 are located in the edge of the water absorption wavelength range, and bands 14–18 fall in the absorption

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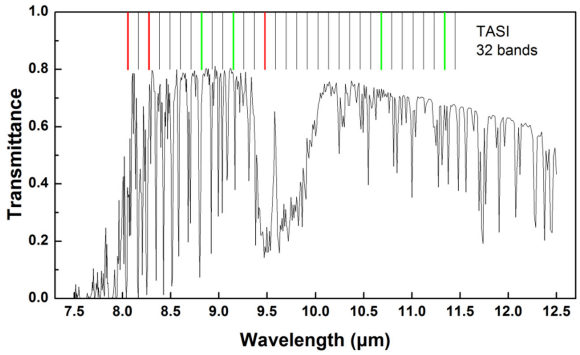


Fig. 1. Locations of total 32 bands and the seven feature bands. The background is the atmospheric transmittance.

ranges of ozone. The remaining bands are mainly in the atmospheric window that have high transmittance. For the simultaneous retrieval of atmospheric and land surface parameters, the seven selected feature bands should contain the atmospheric window bands that have many ground information and nonatmospheric window bands that have much atmospheric information. Much attention should be paid to the water vapor because it has more influence on the TIR signal in the atmosphere [46]. The selected bands are displayed with red and green bars in Fig. 1. The red bar bands indicate the atmospheric absorption bands, where bands 1 (8.055 $\mu$m) and 3 (8.274 $\mu$m) are for water vapor absorption, band 14 (9.478 $\mu$m) indicates ozone absorption, and the green bar bands (8, 11, 25, and 31) are for land surface information because of the high transmittance.

It needs to know the coefficients ($a_{m,i}$ and $b_i$) in (2) from the atmospheric downward and upward radiance, transmittance, and land surface emissivity of the seven feature bands to those values of the rest bands, respectively. For the atmospheric parameters, the MODTRAN 5.2 model was first used to predict the atmospheric downward, upward radiance, and transmittance in TASI 32 bands (assumed with a Gaussian response filter) in terms of the Thermodynamic Initial Guess Retrieval (TIGR) atmospheric profiles. The TIGR database represents a global set of atmospheric situations from the polar to the tropical atmosphere. Of the 946 available profiles under clear sky conditions, 236 are for tropical atmosphere, 258 are for mid-latitude atmosphere, and 452 are for the polar atmosphere. The water vapor varies from 0.06 to 6.3 g/cm$^2$, and the bottom layer air temperature is from 230 to 310 K. A total of 946 groups of atmospheric parameters were obtained and applied to obtain the coefficients ($a_{m,i}$ and $b_i$) in (2) through least-squares regression. Fig. 2(a)–(c) shows that the root-mean-square errors (RMSEs) caused by (2) for atmospheric downward, upward radiance, and transmittance vary with atmosphere profiles. The RMSEs of atmospheric downward and upward radiance have no obvious difference in tropical, mid-latitude, and polar atmospheres and mainly range from 0.01 to 0.02 W/m$^2$/$\mu$m/sr, which is less than 1% of the total value. The RMSEs of transmittance range from 0.001 to 0.004, which is smaller compared with the total transmittance. However, the RMSEs change obviously with the TASI bands, as shown in Fig. 2(d). The ozone absorption bands (13, 15–18) result in a large error probably because only band 14 in this wavelength range is selected as the feature band and cannot reflect the absorption and transmittance features in other bands. Therefore, the number of feature bands should be added in theory. As stated above, the number of feature bands is limited to seven, and many absorption bands correspond to a few nonabsorption bands. This condition consequently increases the error in nonabsorption bands that are used to retrieve the land surface parameters. Fortunately, the ozone has a small influence on the transmittance of thermal infrared wavelengths, and the error in Fig. 2(d) does not cause much uncertainty to the result. Thus, the above seven feature bands were not changed and used for further algorithm development.

For the emissivity, the emissivity spectra of 77 samples from the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) [47] and University of California, Santa Barbara (UCSB) emissivity database [48], including 14 leaf samples, 36 soil samples, 23 rock samples, and 4 water samples, were used. The emissivity of TASI 32 bands was calculated using the band spectral response function by the following equation:

$$\varepsilon = \frac{\int_{\lambda 1}^{\lambda 2} f(\lambda)\varepsilon(\lambda)d\lambda}{\int_{\lambda 1}^{\lambda 2} f(\lambda)d\lambda} \tag{3}$$

where $f(\lambda)$ is the band spectral response function and $\varepsilon(\lambda)$ is the emissivity of one sample at wavelength $\lambda$.

Emissivities of TASI bands of the 77 selected samples are shown in Fig. 3. The spectral contrast of rock samples is the largest, and their maximum and minimum emissivity of the TASI band are, respectively, 0.984 and 0.702, followed by soil and vegetation with the emissivity range from 0.733 to 0.994 and 0.902 to 0.992, respectively. The emissivity of different water samples varies a little with the TASI bands, and the range of emissivity is from 0.968 to 0.994.

Consequently, the coefficients ($a_{m,i}$, and $b_i$) in (2) were regressed. As shown in Fig. 4(a), the RMSEs of emissivity varying with the number of samples number are mostly smaller than 0.006 except for some rock samples with high emissivity variability. Due to the differences of emissivity distribution of various samples shown in Fig. 3, the regression residuals using (2) present similar patterns, and the errors of a few soil and vegetation samples are higher than the average value of the rest of the same kind of samples. Besides, the RMSEs of emissivity in TASI bands are less than 0.005 [Fig. 4(b)]. Therefore, emissivity regression residuals do not cause obvious errors in the retrieval results. The above results obtained from Figs. 2 and 4 indicate that the linear empirical model of (2) can be used to predict the atmospheric downward and upward radiance, transmittance, and land surface emissivity in nonfeature bands from those values of the feature bands. This condition shows that the number of unknowns is reduced effectively by (2).

### *C. Initial Guess for the Unknowns Using DNN*

The DNN model is used to obtain the initial values ($T_s^{(0)}$, $\varepsilon^{(0)}$, $L_a^{\downarrow(0)}$, $L_a^{\uparrow(0)}$, and $\tau^{(0)}$). Considering the region limitation of ground observations, with the above atmospheric upward radiance, downward radiance, and transmittance simulated from the MODTRAN model with 258 mid-latitude TIGR

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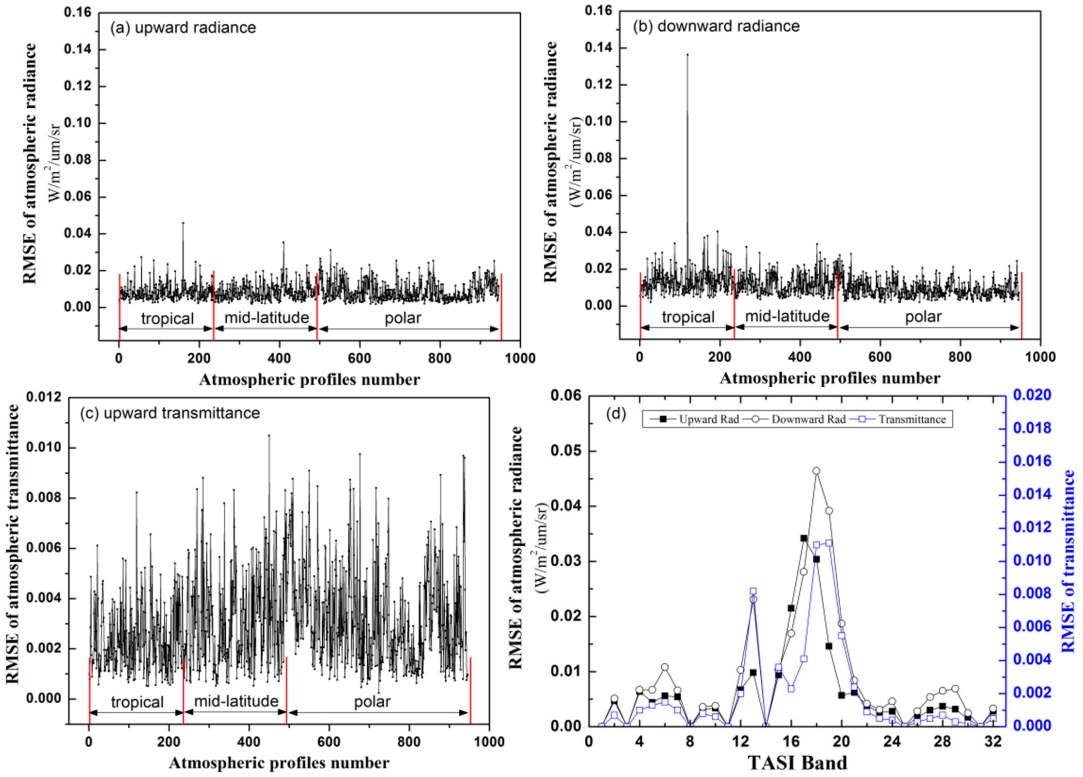


Fig. 2. RMSEs of (a) atmospheric upward radiance, (b) atmospheric downward radiance, and (c) atmospheric upward transmittance for different profiles and for the (d) TASI bands.

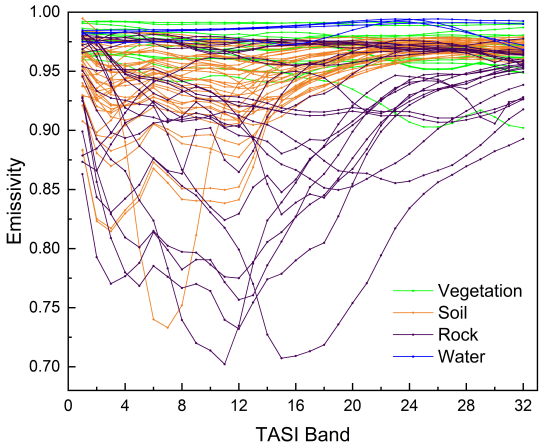


Fig. 3. Emissivities of TASI bands of the selected samples.

atmospheric profiles, we used surface temperature $T_s$ and emissivity to derive the thermal radiative transfer equation [see (1)] to simulate the apparent radiance of TASI 32 bands. A total of 16 LSTs $T_s$ for each atmospheric profile were used, with $T_s = T_0 - 10$ K to $T_0 + 20$ K at an interval of 2 K, where $T_0$ is the bottom air temperature of the specific atmospheric profile. The band apparent radiance was simulated for a total of 317 856 cases, including 258 mid-latitude atmospheric profiles, 77 emissivity samples, and 16 levels of surface temperatures $T_s$. In the simulation data set, LST $T_s$ varies from 231.25 to 318.65 K. For atmospheric parameters, the ranges of column water vapor, atmospheric transmittance, downward radiance, and upward radiance are [0, 2.8] g/cm$^2$, [0.651, 0.995], [0.118, 5.272] W/m$^2$/$\mu$m/sr, and [0.024, 1.927] W/m$^2$/$\mu$m/sr, respectively. Since the real TIR remote sensing data may be contaminated by instrument measurement noise, a noise-equivalent-change-in-temperature (NE$\Delta$T, designed as 0.2 K for TASI) was added to the simulated brightness temperature in each band. The DNN model is then applied to train the simulation data set with the apparent radiance of TASI 32 TIR bands as input and the 29 unknowns, including $\varepsilon_i$, $L_{ai}^{\downarrow}$, $L_{ai}^{\uparrow}$, and $\tau_i$ in seven feature bands and $T_s$, as mentioned in Section II-B. The trained DNN model is then applied to obtain the initial values of unknowns from each pixel of the TASI image, and the initial values are then used in the physical model to obtain the final retrieval results.

### *D. Physical Retrieval Model Development*

Although the above linear empirical model reduces the number of unknowns, the retrieval of the 29 parameters remains difficult because $T_s$, $\varepsilon$, $L_a^{\downarrow}$, $L_a^{\uparrow}$, and $\tau$ are coupled with each other. Therefore, the radiative transfer equation [see (1)] should be modified. Given that a group of initial values is known as $T_s^{(0)}$, $\varepsilon^{(0)}$, $L_a^{\downarrow(0)}$, $L_a^{\uparrow(0)}$, and $\tau^{(0)}$, (1) can be written as (3). For simplicity, we removed subscript $i$

$$L^{(0)} = \left[\varepsilon^{(0)} B\left(T_s^{(0)}\right) + (1-\varepsilon^{(0)}) L_a^{\downarrow(0)}\right] \cdot \tau^{(0)} + L_a^{\uparrow(0)}. \quad (4)$$

Consequently, the difference between (1) and (4) is expressed as

$$\begin{aligned} L - L^{(0)} = & \underbrace{\varepsilon B(T_s)\tau - \varepsilon^{(0)} B(T_s^{(0)})\tau^{(0)}}_{\text{residual of surface emission}} \\ & + \underbrace{(1-\varepsilon)L_a^{\downarrow} + \tau(1-\varepsilon^{(0)})L_a^{\downarrow(0)}\tau^{(0)}}_{\text{residual of reflected downward atmospheric radiance}} \\ & + \underbrace{L_a^{\uparrow} - L_a^{\uparrow(0)}}_{\text{residual of upward atmospheric radiance}}. \end{aligned} \quad (5)$$

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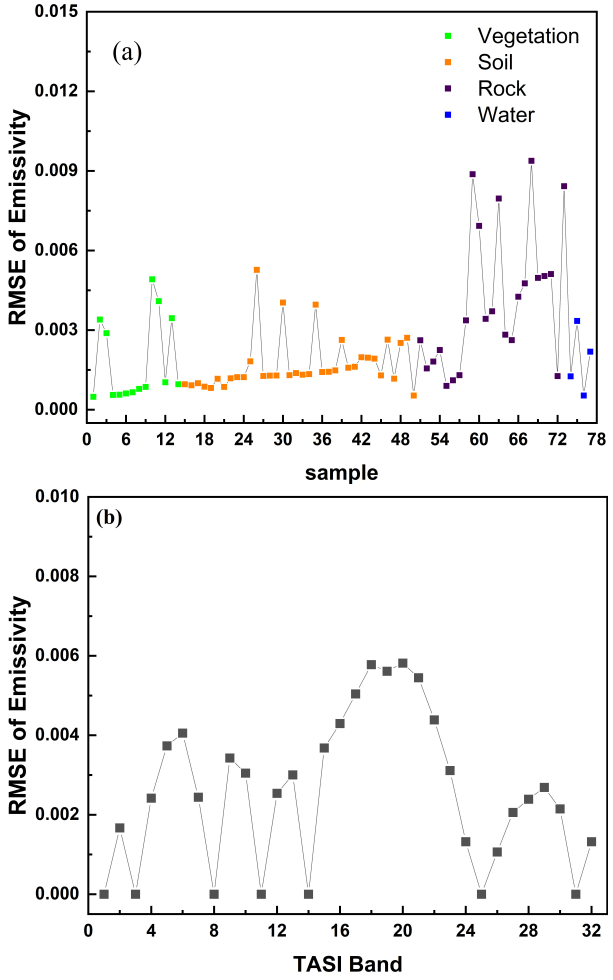


Fig. 4. RMSEs of emissivity for (a) different samples and (b) TASI bands.

The first term of the right side of (5) is the residual of surface emission caused by the initial value, the second term is the residual of surface reflected atmospheric downward radiance, and the third term is the residual of atmospheric upward radiance. Equation (5) can be explained as the first-order variation of (1) at the initial values of $T_s^{(0)}$, $\varepsilon^{(0)}$, $L_a^{\downarrow(0)}$, $L_a^{\uparrow(0)}$, and $\tau^{(0)}$. Based on (1), the terms in the thermal radiative transfer equation are independent of each other, and the first-order variation of this equation can be approximated as follows:

$$\begin{aligned}\delta L &= \varepsilon B\delta\tau + B\tau\delta\varepsilon + \varepsilon\tau\delta B + (1-\varepsilon)\tau\delta L_a^{\downarrow} \\ &\quad + (1-\varepsilon)L_a^{\downarrow}\delta\tau - L_a^{\downarrow}\tau\delta\varepsilon + \delta L_a^{\uparrow} \\ &= \varepsilon\tau\delta B + (B\tau - L_a^{\downarrow}\tau)\delta\varepsilon + \left[\varepsilon B - (1-\varepsilon)L_a^{\downarrow}\right]\delta\tau \\ &\quad + (1-\varepsilon)\tau\delta L_a^{\downarrow} + \delta L_a^{\uparrow}. \end{aligned} \tag{6}$$

Equation (6) is valuable because it changes the nonlinear expression of the radiative transfer equation in (1) to a linear format, with a group of initial values that are required to be closed to the final truth. Thus, the retrieval of $T_s$, $\varepsilon$, $L_a^{\downarrow}$, $L_a^{\uparrow}$, and $\tau$ changes to estimating their residual values $\delta B$, $\delta\varepsilon$, $\delta\tau$, $\delta L_a^{\uparrow}$, and $\delta L_a^{\downarrow}$, as $T_s = T_s^{(0)} + \delta T_s$, $\varepsilon = \varepsilon_s^{(0)} + \delta\varepsilon$, $\tau = \tau_s^0 + \delta\tau$, $L_a^{\downarrow} = L_a^{\downarrow(0)} + \delta L_a^{\downarrow}$, and $L_a^{\uparrow} = L_a^{\uparrow(0)} + \delta L_a^{\uparrow}$.

Moreover, the band radiance ($L$ and $B$) can be approximated as a quadratic expression of temperature, which can be expressed as

$$L \approx g \cdot T_{bn}^2 + h \cdot T_{bn} + k \text{ and } B \approx g \cdot T_s^2 + h \cdot T_s + k. \tag{7}$$

The errors of the quadratic approximation are very small in the entire LST range of the simulation data set (231.25–318.65 K), and the interval of the RMSEs between the real radiance and the approximated radiance of all TASI bands is [0.0038, 0.1030] W/m$^2$/$\mu$m/sr. At the temperature of 230, 260, 290, and 320 K, the approximation errors of the first TASI band are only 0.0104, 0.1042, −0.0679, and −0.0850 W/m$^2$/$\mu$m/sr, respectively. As a result, the quadratic approximation will not bring obvious retrieval errors. And their first derivatives are

$$\delta L \approx (2g \cdot T_{bn} + h)\delta T_{bn}, \text{ and } \delta B \approx (2g \cdot T_s + h)\delta T_s \tag{8}$$

where $T_{bn}$ and $T_s$ are the brightness temperatures at observation height and LST, respectively, and $g$, $h$, and $k$ are the coefficients that vary with different TASI bands. Using (6) to (8) obtains a new expression

$$\begin{aligned}&(2gT_{bn} + h)\delta T_{bn} \\ &\quad = \varepsilon\tau(2gT_s + h)\delta T_s + \left(B\tau - L_a^{\downarrow}\tau\right)\delta\varepsilon \\ &\qquad + \left[\varepsilon B - (1-\varepsilon)L_a^{\downarrow}\right]\delta\tau + (1-\varepsilon)\tau\delta L_a^{\downarrow} + \delta L_a^{\uparrow}. \end{aligned} \tag{9}$$

That is

$$\begin{aligned}\delta T_{bn} &= \frac{1}{2gT_{bn} + h}\left\{\varepsilon\tau(2gT_s + h)\delta T_s + \left(B\tau - L_a^{\downarrow}\tau\right)\delta\varepsilon \right. \\ &\qquad \left. + \left[\varepsilon B - (1-\varepsilon)L_a^{\downarrow}\right]\delta\tau \right. \\ &\qquad \left. + (1-\varepsilon)\tau\delta L_a^{\downarrow} + \delta L_a^{\uparrow}\right\} \\ &= W_T \cdot \delta T_s + W_\varepsilon \cdot \delta\varepsilon + W_\tau \cdot \delta\tau + W_{L^{\downarrow}} \cdot \delta L_a^{\downarrow} + W_{L^{\uparrow}} \cdot \delta L_a^{\uparrow} \end{aligned} \tag{10}$$

with $W_T = (2gT_s + h/2gT_{bn} + h)\varepsilon\tau$, $W_\varepsilon = ((B\tau - L_a^{\downarrow}\tau)/2gT_{bn} + h)$, $W_\tau = (\varepsilon B - (1-\varepsilon)L_a^{\downarrow}/2gT_{bn} + h)$, $W_{L^{\downarrow}} = ((1-\varepsilon)\tau/2gT_{bn} + h)$, and $W_{L^{\uparrow}} = (1/2gT_{bn} + h)$. All of them are the coefficients, and they are calculated in each update starting from the above initial values ($T_s^{(0)}$, $\varepsilon^{(0)}$, $L_a^{\downarrow(0)}$, $L_a^{\uparrow(0)}$, and $\tau^{(0)}$).

Equation (10) can be written as a matrix

$$\delta T_{bn} = w \times x \tag{11}$$

with $w = (W_T, W_\varepsilon, W_\tau, W_{L^{\downarrow}}, W_{L^{\uparrow}})$ and $x = (\delta T_s, \delta\varepsilon, \delta\tau, \delta L_a^{\downarrow}, \delta L_a^{\uparrow})^{\mathrm{T}}$. However, (11) only presents the case of one band. For the 32 TIR bands of TASI, 29 unknowns are found in the seven feature bands after the empirical models shown in (2). Consequently, vector $x$ in (11) is expressed in the following form:

$$\begin{aligned}X = \big(&\delta T_s, \delta\varepsilon_1, \ldots, \delta\varepsilon_7, \delta\tau_1 \ldots \delta\tau_7, \delta L_{a1}^{\downarrow}, \\ &\ldots, \delta L_{a7}^{\downarrow}, \delta L_{a1}^{\uparrow}, \ldots, \delta L_{a7}^{\uparrow}\big)^{\mathrm{T}}. \end{aligned} \tag{12}$$

For the 32 bands, the left side of (11) can be expressed as (13). For better understanding, (13) rearranges the order of TASI bands by placing the seven feature bands forward. Subscripts 1, 2, 3, ..., 7 present the seven feature bands, and 8, 9, ..., 32 are the remaining bands

$$Y = \left(\delta T_{bn1}, \delta T_{bn2}, \delta T_{bn3} \ldots \delta T_{bn7}, \delta T_{bn8}, \delta T_{bn9}, \ldots, \delta T_{bn32}\right)^{\mathrm{T}}. \tag{13}$$

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Considering the empirical model in (2), $w$ in (11) for the 32 bands becomes the coefficient matrix in (14), as shown at the bottom of the page. In this matrix, the first seven rows are the coefficients of the seven feature bands, and the remaining rows are for nonfeature bands. The terms $a_{j,i}$ ($j = 8, 9, \ldots, 32$, and $i = 1, 2, \ldots, 7$) are the coefficients used to convert the seven feature bands' emissivity to that of the $i$th band, similar to $a_{m,i}$ in (2). Similarly, $b_{j,i}$, $c_{j,i}$, and $d_{j,i}$ are the coefficients for atmospheric transmittance, downward radiance, and upward radiance, respectively.

Combining (11)–(14) obtains the final radiative transfer equations, which are written in the matrix as

$$Y = W \times X \tag{15}$$

where $Y$ is a $32 \times 1$ vector related to the observed data [see (13)], $X$ is a $29 \times 1$ vector related to the parameters required for retrieval [see (12)], and $W$ is a $32 \times 29$ coefficient matrix [see (14)]. The residual is obtained as

$$\Delta T_b = \|WX - Y\|_2^2 + \gamma^2 \|LX\|_2^2. \tag{16}$$

The first term of (16) is the residual of (15), and the second term denotes the boundary constraints similar to the effective value range in Section II-B.

Borel [49] found that the correct land surface emissivity spectrum is the smoothest because it eliminates the effects of atmospheric radiance, and the smoothness is computed as

$$\sigma(\varepsilon_n) = \mathrm{STDEV}\left(\varepsilon_{n,m} - \frac{\varepsilon_{n,m-1} + \varepsilon_{n,m} + \varepsilon_{n,m+1}}{3}\right) \tag{17}$$

where $\sigma(\varepsilon_n)$ is the standard deviation of the $n$th emissivity spectrum, and $\varepsilon_{n,m}$ is the $n$th emissivity of the $m$th sample.

Therefore, the smoothness is considered in this study, and the cost function is composed of (16) and (17) as

$$Loss = w_1 \cdot \Delta T_b + w_2 \cdot \sigma(\varepsilon) \rightarrow \min \tag{18}$$

and the final solution is reached when the minimum loss function is obtained by the following steps.

1) Based on the observed radiance $L$ in 32 bands and the initial values ($T_s^{(0)}$, $\varepsilon^{(0)}$, $L_a^{\downarrow(0)}$, $L_a^{\uparrow(0)}$, and $\tau^{(0)}$) from the DNN model, the error of brightness temperature and the coefficient matrix are calculated by (6)–(10).
2) The residual vector $X$ in (12) is initialized and substituted into (14)– (17) to calculate the loss function in (18). Then, the value of $X$ will be optimized iteratively to reduce the value of the loss function. In this study, $X$ is optimized by the Levenberg–Marquardt algorithm [50], which is a commonly used and robust least-squares optimization algorithm.
3) The physical model results ($T_s^{\mathrm{phy}}$, $\varepsilon^{\mathrm{phy}}$, $L_a^{\downarrow phy}$, $L_a^{\uparrow phy}$, and $\tau^{\mathrm{phy}}$) are obtained after the loss function minimization is reached or the number of iterations reaches the maximum, which consists of optimized residual vector $X^{\mathrm{opt}} = (\delta T_s^{\mathrm{opt}}, \delta\varepsilon_1^{\mathrm{opt}}, \ldots, \delta\varepsilon_7^{\mathrm{opt}}, \delta\tau_1^{\mathrm{opt}} \ldots \delta\tau_7^{\mathrm{opt}}, \delta L_{a1}^{\downarrow opt}, \ldots, \delta L_{a7}^{\downarrow opt}, \delta L_{a1}^{\uparrow opt}, \ldots, \delta L_{a7}^{\uparrow opt})^{\mathrm{T}}$ and initial retrieval values. For example, the final retrieved LST is calculated as

$$T_s^{\mathrm{phy}} = T_s^0 + \delta T_s^{\mathrm{opt}}. \tag{19}$$

## III. Algorithm Analysis

### A. Initial Values Analysis From the DNN Model

A multilayer perceptron DNN model with six layers, including one input layer, four hidden layers, and one output layer, was designed to obtain the initial land surface and atmospheric parameter values from TASI 32-band radiances. The number of neurons in each hidden layer is 64, 64, 48, and 48. Rectified linear unit (ReLU) activation function and Adam optimization method were used in our DNN model, and the loss function is the mean squared error (MSE) of the real values and the retrieved $\varepsilon_i$, $L_{ai}^{\downarrow}$, $L_{ai}^{\uparrow}$, $\tau_i$ in seven feature bands and $T_s$. The simulation data set, described in Section II-B, was divided into a training set with 90% data and a testing set with 10% data randomly chosen from each atmospheric and land condition. The training set was used to train the DNN model, and the testing set was used to evaluate the model performance. The LST residual and RMSEs of emissivity and atmospheric parameters of each band are shown in Fig. 5(a)–(e), separately. Following the histogram of LST residuals in Fig. 5(a), the overall RMSE and bias are 0.98 and −0.23 K, respectively, and the histogram conforms to a normal distribution with a mean value approximately equal to zero. Fig. 5(b) shows that the emissivity RMSEs of most TASI bands do not exceed 0.016, except for the 15th band (9.588 $\mu$m). For atmospheric parameters, their error trends are essentially the same. The largest errors are in bands 1 and 2, with maximum RMSEs of 0.0331, 0.2488, and 0.5919 W/m$^2$/$\mu$m/sr for transmittance, upward radiance, and downward radiance, respectively. The RMSEs decrease until bands 15–20, with minimum RMSEs of 0.008, 0.0623, and 0.2698 W/m$^2$/$\mu$m/sr, and the RMSE of each atmospheric parameter slightly increases until band 31. In general, the DNN model achieves estimation results with high accuracy and is suitable to be used as the initial values of the physical model.

$$W = \begin{bmatrix}
W_{T1} & W_{\varepsilon 1}\ 0\ 0\ 0\ 0\ 0\ 0 & W_{\tau 1}\ 0\ 0\ 0\ 0\ 0\ 0 & W_{L\downarrow 1}\ 0\ 0\ 0\ 0\ 0\ 0 & W_{L\uparrow 1}\ 0\ 0\ 0\ 0\ 0\ 0 \\
W_{T2} & 0\ W_{\varepsilon 2}\ 0\ 0\ 0\ 0\ 0 & 0\ W_{\tau 2}\ 0\ 0\ 0\ 0\ 0 & 0\ W_{L\downarrow 2}\ 0\ 0\ 0\ 0\ 0 & 0\ W_{L\uparrow 2}\ 0\ 0\ 0\ 0\ 0 \\
W_{T3} & 0\ 0\ W_{\varepsilon 3}\ 0\ 0\ 0\ 0 & 0\ 0\ W_{\tau 3}\ 0\ 0\ 0\ 0 & 0\ 0\ W_{L\downarrow 3}\ 0\ 0\ 0\ 0 & 0\ 0\ W_{L\uparrow 3}\ 0\ 0\ 0\ 0 \\
\cdots & \cdots & \cdots & \cdots & \cdots \\
W_{T7} & 0\ 0\ 0\ 0\ 0\ 0\ W_{\varepsilon 7} & 0\ 0\ 0\ 0\ 0\ 0\ W_{\tau 7} & 0\ 0\ 0\ 0\ 0\ 0\ W_{L\downarrow 7} & 0\ 0\ 0\ 0\ 0\ 0\ W_{L\uparrow 7} \\
W_{T8} & W_{\varepsilon 8}(a_{8,1}a_{8,2}a_{8,3}a_{8,4}a_{8,5}a_{8,6}a_{8,7}) & W_{\tau 8}(b_{8,1}b_{8,2}b_{8,3}b_{8,4}b_{8,5}b_{8,6}b_{8,7}) & W_{L\downarrow 8}(c_{8,1}c_{8,2}c_{8,3}c_{8,4}c_{8,5}c_{8,6}c_{8,7}) & W_{L\uparrow 8}(d_{8,1}d_{8,2}d_{8,3}d_{8,4}d_{8,5}d_{8,6}d_{8,7}) \\
W_{T9} & W_{\varepsilon 9}(a_{9,1}a_{9,2}a_{9,3}a_{9,4}a_{9,5}a_{9,6}a_{9,7}) & W_{\tau 9}(b_{9,1}b_{9,2}b_{9,3}b_{9,4}b_{9,5}b_{9,6}b_{9,7}) & W_{L\downarrow 9}(c_{9,1}c_{9,2}c_{9,3}c_{9,4}c_{9,5}c_{9,6}c_{9,7}) & W_{L\uparrow 9}(d_{9,1}d_{9,2}d_{9,3}d_{9,4}d_{9,5}d_{9,6}d_{9,7}) \\
\cdots & \cdots & \cdots & \cdots & \cdots \\
W_{T32} & W_{\varepsilon 32}(a_{32,1}a_{32,2}a_{32,3}a_{32,4}a_{32,5}a_{32,6}a_{32,7}) & W_{\tau 32}(b_{32,1}b_{32,2}b_{32,3}b_{32,4}b_{32,5}b_{32,6}b_{32,7}) & W_{L\downarrow 32}(c_{32,1}c_{32,2}c_{32,3}c_{32,4}c_{32,5}c_{32,6}c_{32,7}) & W_{L\uparrow 32}(d_{32,1}d_{32,2}d_{32,3}d_{32,4}d_{32,5}d_{32,6}d_{32,7})
\end{bmatrix} \tag{14}$$

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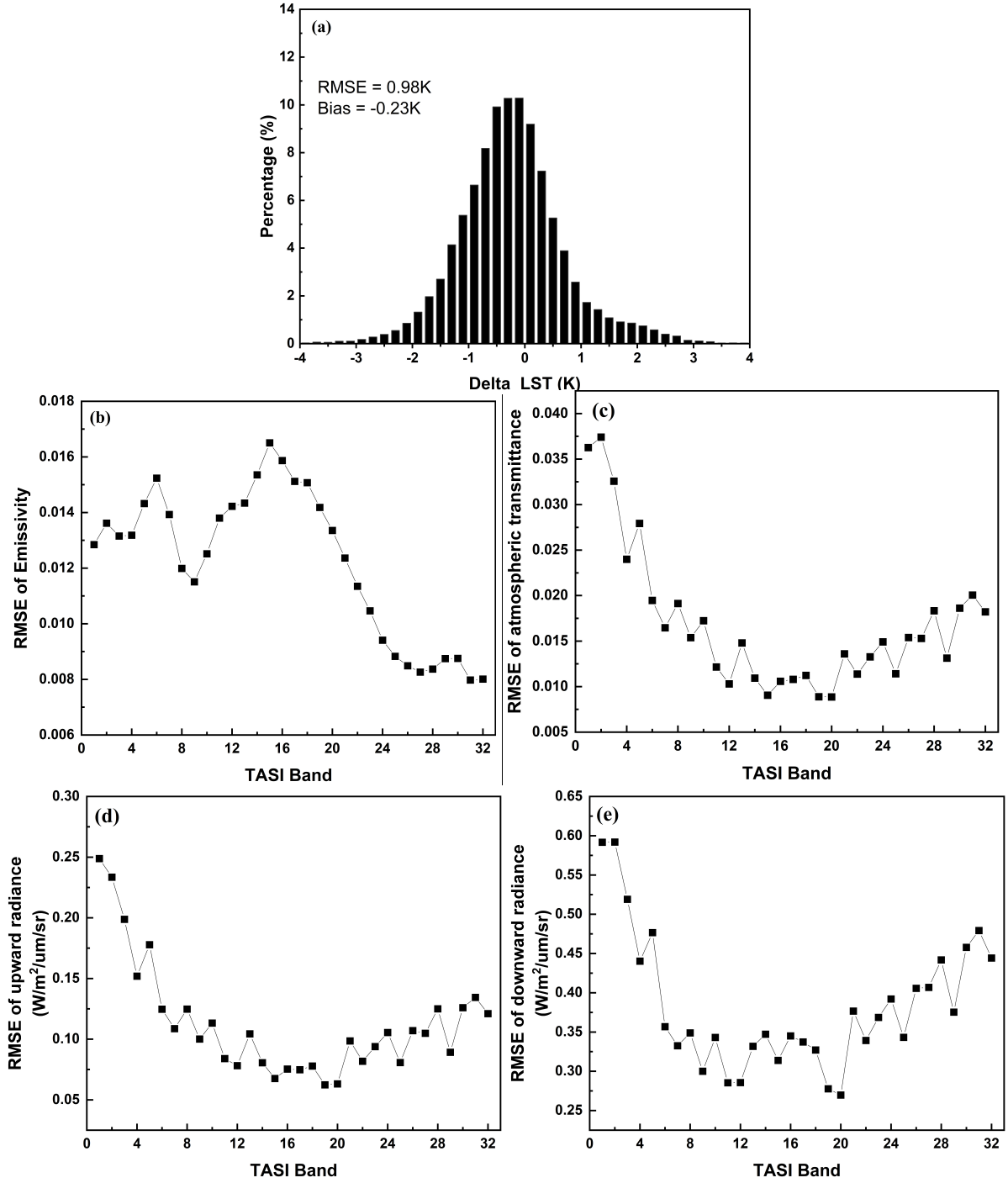


Fig. 5. Error evaluation results of (a) LST, (b) emissivity, (c) atmospheric transmittance, (d) atmospheric upward radiance, and (e) atmospheric downward radiance from the DNN model.

*B. Retrieval Results Analysis From the Physical Model*

With the initial values from the DNN model, the physical model gives the final retrieval results, and its errors of different parameters are shown in Fig. 6. As shown in Fig. 6(a), the LST residual is effectively corrected, and the overall RMSE reduces from 0.98 to 0.86 K. The mean value of LST retrieval error is close to 0, with bias changing from −0.23 to 0.05 K compared with those of Fig. 5(a). For land surface emissivity, the accuracy also significantly improves after the optimization of the proposed physical model. The RMSE of each band is lower than that of the DNN model, and the maximum RMSE is reduced to 0.0129 from greater than 0.015 for bands 14–18 in the DNN model results. The average emissivity RMSE of those bands is reduced by more than 0.003. For atmospheric parameters, the errors of most bands decrease, except for the downward radiance of bands 16–19. This finding is probably due to the linear fitting error of (2). It should be also noted that the RMSEs of the retrieved atmospheric parameters of bands 1–6 are larger than the rest. The possible reasons include the following.

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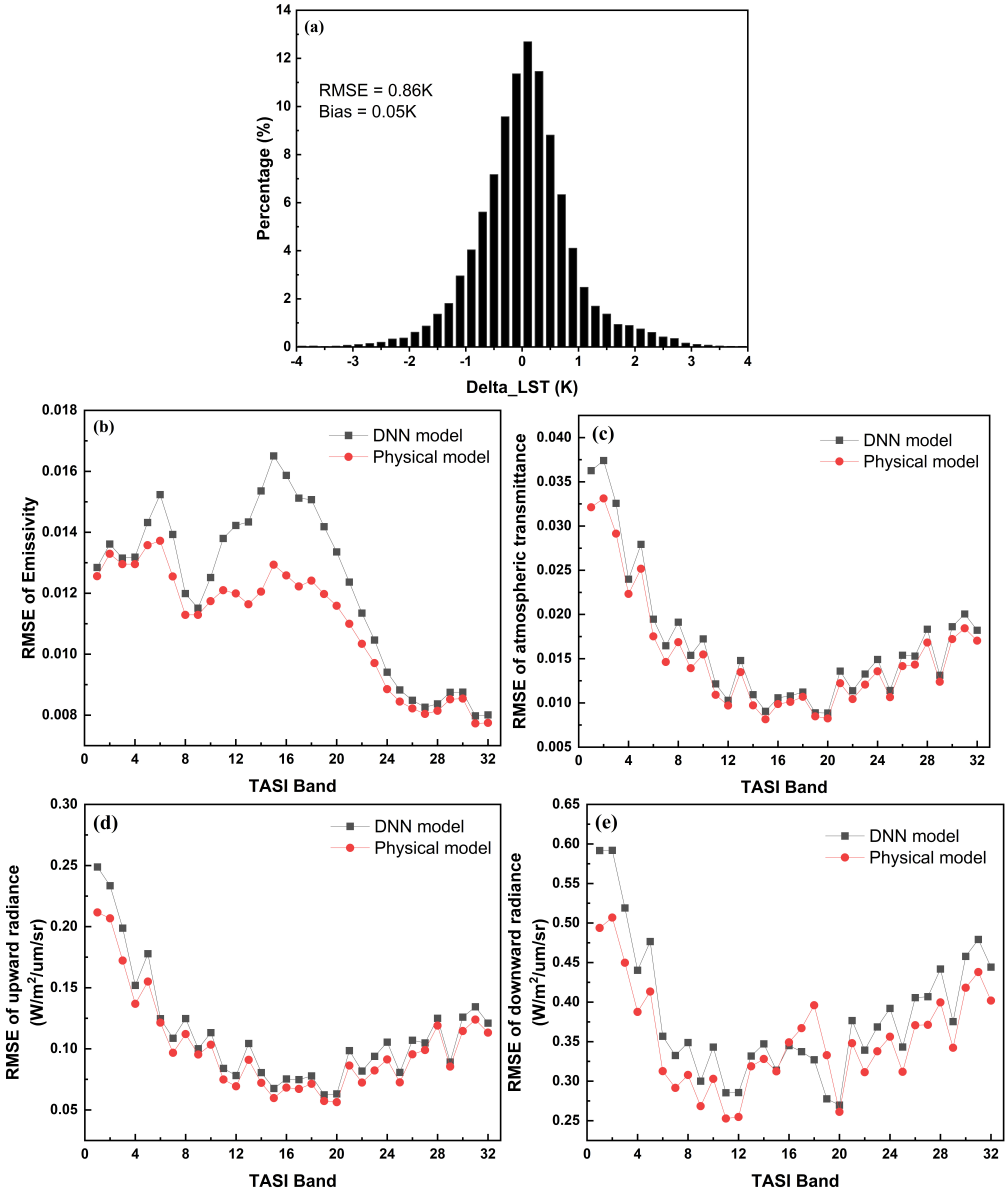


Fig. 6. Error results of (a) LST, (b) emissivity, (c) atmospheric transmittance, (d) atmospheric upward radiance, and (e) atmospheric downward radiance from the physical model and comparison with the results of the DNN model.

1) There is a high variability of the atmospheric transmittance of bands 1–6 which can be seen from Fig. 1, and it may bring more uncertainty.
2) In the simulation data set, an NE$\Delta$T of 0.2 K is added to each band, but the observed radiance of bands 1–6 is lower than that of other bands based on Planck's law; therefore, the error ratio of these bands is consequently larger.
3) In addition, it is found that the quadratic approximation error of the Planck's law in these bands is higher than in other bands, and it can also lead to relatively higher uncertainty.

The shape of the emissivity curve is an important indicator of evaluation. In this study, the discrete Fréchet distance [51] was used to measure the similarity between the real and retrieval emissivity curves. The smaller the value of discrete Fréchet distance is, the more similar the emissivity curve shape will be. The discrete Fréchet distance is calculated as

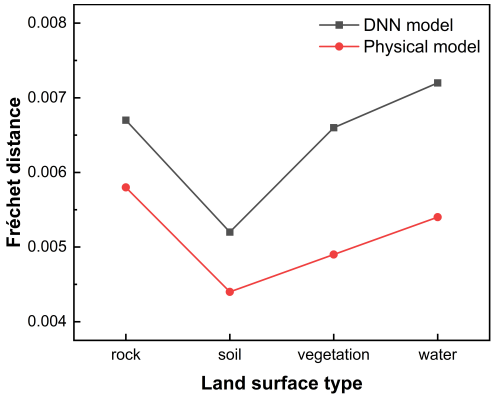


Fig. 7. Emissivity curve similarity assessment for the DNN and physical models.

$$\delta_{dF}(P, Q) = \min\{\| \max_{i=1,\ldots,m} d(u_{ai}, v_{bi}) \|\} \tag{20}$$

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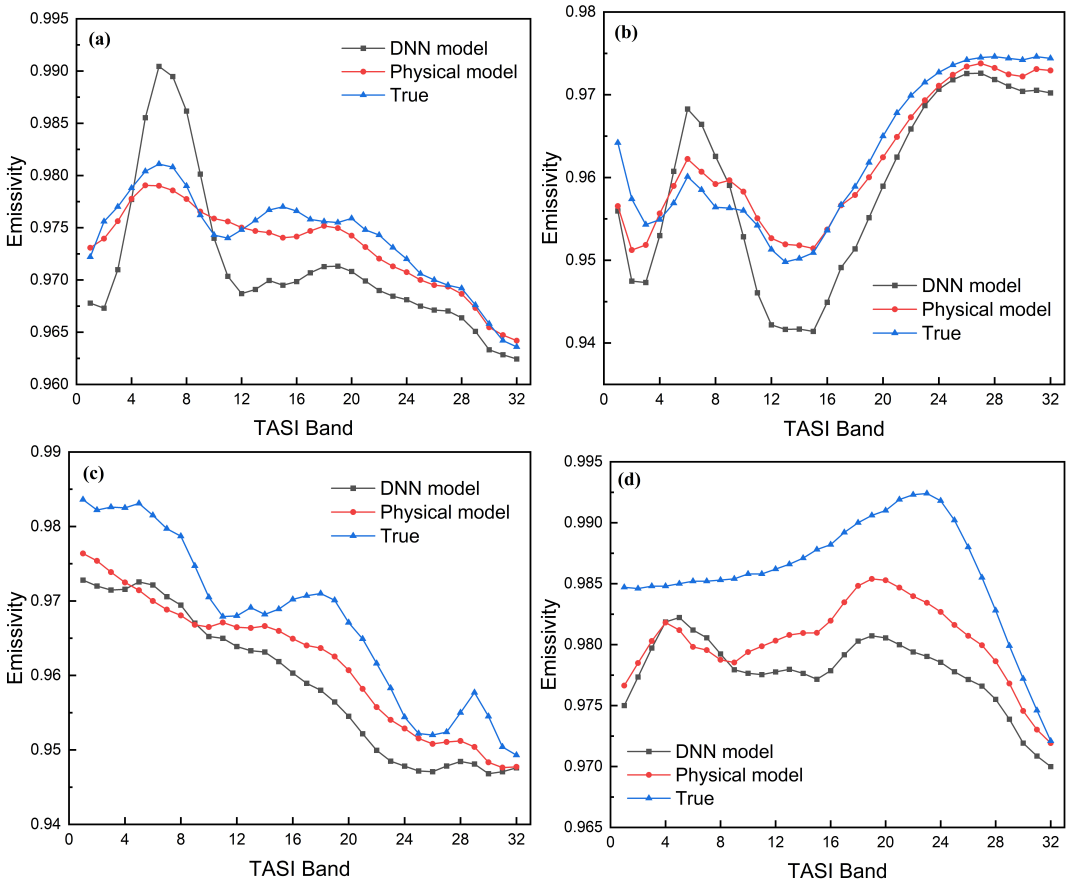


Fig. 8. Comparison of emissivity curves for (a) rock, (b) soil, (c) vegetation, and (d) water.

where $P$ and $Q$ are polygonal curves; $\sigma(P) = (u_1, \ldots, u_p)$ and $\sigma(Q) = (v_1, \ldots, v_p)$ are the corresponding sequences. A coupling between $P$ and $Q$ is a sequence of distinct pairs from $\sigma(P) \times \sigma(Q)$ as $(u_{a1}, v_{b1}), (u_{a2}, v_{b2}), \ldots, (u_{am}, v_{bm})$, and $d$ is a distance function, such as Euclidean distance.

To evaluate the emissivity results of the physical model in detail, separate statistics is performed for different land surface types. The results are shown in Fig. 7. Compared with the results of the DNN model, the emissivity curve of each land surface type in the physical model results is significantly closer to the real value. Some examples of emissivity curve comparison for each land surface type are shown in Fig. 8. The result shows that the physical model results are closer to the real values in different cases. Therefore, it can be concluded that the physical model effectively improves retrieval accuracy than the DNN model.

## IV. Application and Validation

### *A. Retrieval of LST and Emissivity From Airborne TASI Image*

For simplification, the proposed algorithm is called the feature-band linear-format hybrid algorithm (FebLihy algorithm). According to the above discussion, Fig. 9 shows the proceeding to retrieve LST, emissivity, and atmospheric data from the TASI image. First, the initial values of land surface and atmospheric parameters of feature bands are estimated by the DNN model. Then, it is substituted into the linear radiative

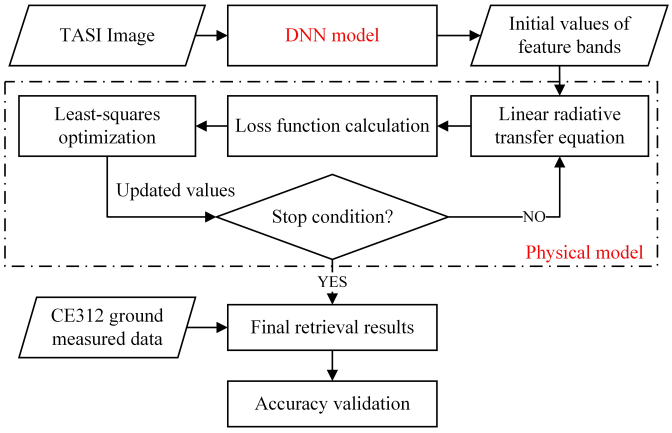


Fig. 9. Flowchart of the FebLihy algorithm.

transfer equation in (1) and updated by the least-squares optimization method based on loss function in (18). The process is iterated until the end condition is satisfied as stated in Section II-D. The final retrieval results are obtained from the physical model, the land surface emissivity is measured with a multiple-band thermal radiometer CE312, and used for accuracy validation.

The FebLihy algorithm was applied in a real TASI image of Fuyun County in Xinjiang Province, China. This area is rich in mineral resources and has a wide variety of landforms, including mountains, basin, river valley, Gobi, and desert. The flight altitude was 1500 m and the TASI image was observed with a spatial resolution of 2 m at UTC 08:04 (approximately

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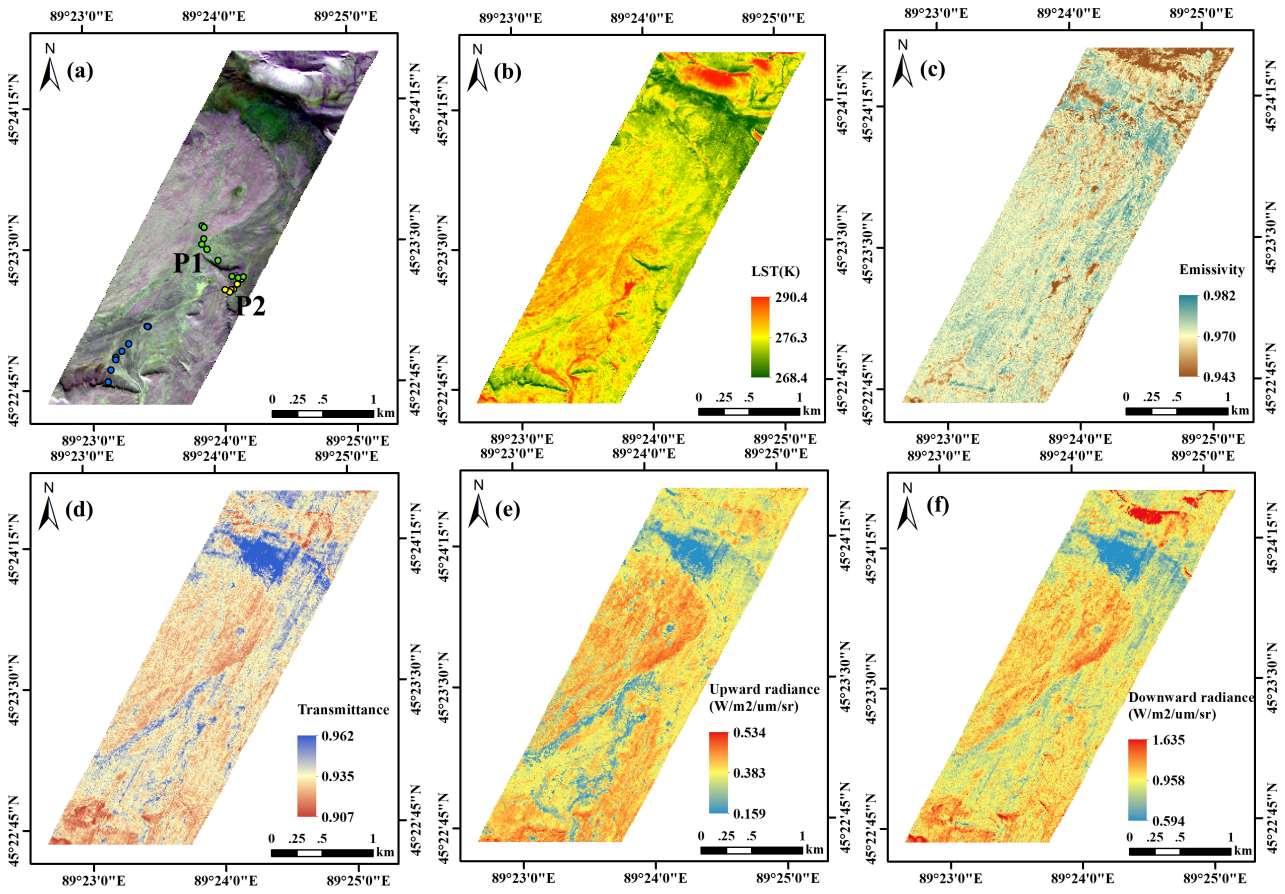


Fig. 10. (a) Original TASI image and (b) retrieval results of LST, (c) emissivity of band 25, (d) atmospheric transmittance of band 25, (e) atmospheric upward radiance of band 25, and (f) atmospheric downward radiance of band 25.

14:00 in local time) on October 26, 2018. The experimental image can be approximately regarded as observed at nadir view with a small angular variation of LST and emissivity [52], because the maximum viewing zenith angle (VZA) of TASI is 20°. Fig. 10(a) shows the false-color image of the study area with R: TASI band 1, G: TASI band 11, and B: TASI band 25. Three different color points in Fig. 10(a) represent the ground measurement locations using a multiband thermal radiometer CE312 at different dates. The yellow points were collected on October 26, the blue points were collected on October 27, and the green points were collected on October 28. The surface type of all points was sandy soil. The distribution of thermal radiance in the study area is complex due to the influence of land cover types and topography. The land surface types of the study area are mainly rock sand and bare soil with slightly sparse vegetation. The terrain in the middle is flat, and mountains are found in the south and north. Land surface and atmospheric parameters were retrieved using the proposed FebLihy algorithm. Fig. 10(b)–(f) displays the LST image, land surface emissivity, atmospheric transmittance, atmospheric upward, and downward radiance images of one of the chosen feature bands (band 25). The LST ranges from 268.4 to 290.4 K, and the emissivity ranges from 0.943 to 0.982. The transmittance ranges from 0.962 to 0.907, and the atmospheric upward and downward radiances range from 0.159 to 0.534 W/m$^2$/$\mu$m/sr and from 0.594 to 1.635 W/m$^2$/$\mu$m/sr, respectively. As shown in Fig. 9, the LST of the mountain is lower than that of flat desert, and high- and low-temperature accumulation areas are found in the North. Most of the low emissivity of band 25 is clustered in the North, and the emissivity of different regions varies greatly in spatial range and TIR band. Atmospheric transmittance is high due to the low flight altitude, and the atmospheric parameters are heavily influenced by the land surface. The spatial distribution is similar to that of land surface radiance in Fig. 10(a).

Two sampling points (P1 selected from sampling points on October 28 and P2 selected from sampling points on October 26) in Fig. 10(a) were selected. Their channel emissivity and atmospheric parameters are displayed in Fig. 11. Although the two sampling points are sandy soil, some obvious differences in emissivity value and shape are found between their emissivity curves. It shows that the hyperspectral thermal infrared technology is helpful to distinguish the subtle features of the land surface, thereby contributing to applications, such as land surface classification and mineral detection. However, the atmospheric parameters at the two sampling points are similar, and the minor differences may be due to the variance of the land surface conditions.

### *B. Emissivity Validation Using Ground-Measured Data*

The CE312 radiometer has 5 bands in 8–12 $\mu$m (8.1–8.5, 8.5–8.9, 8.9–9.3, 10.3–11.0, 11.0–11.7 $\mu$m). These bands are wider than TASI's thermal infrared bands and are

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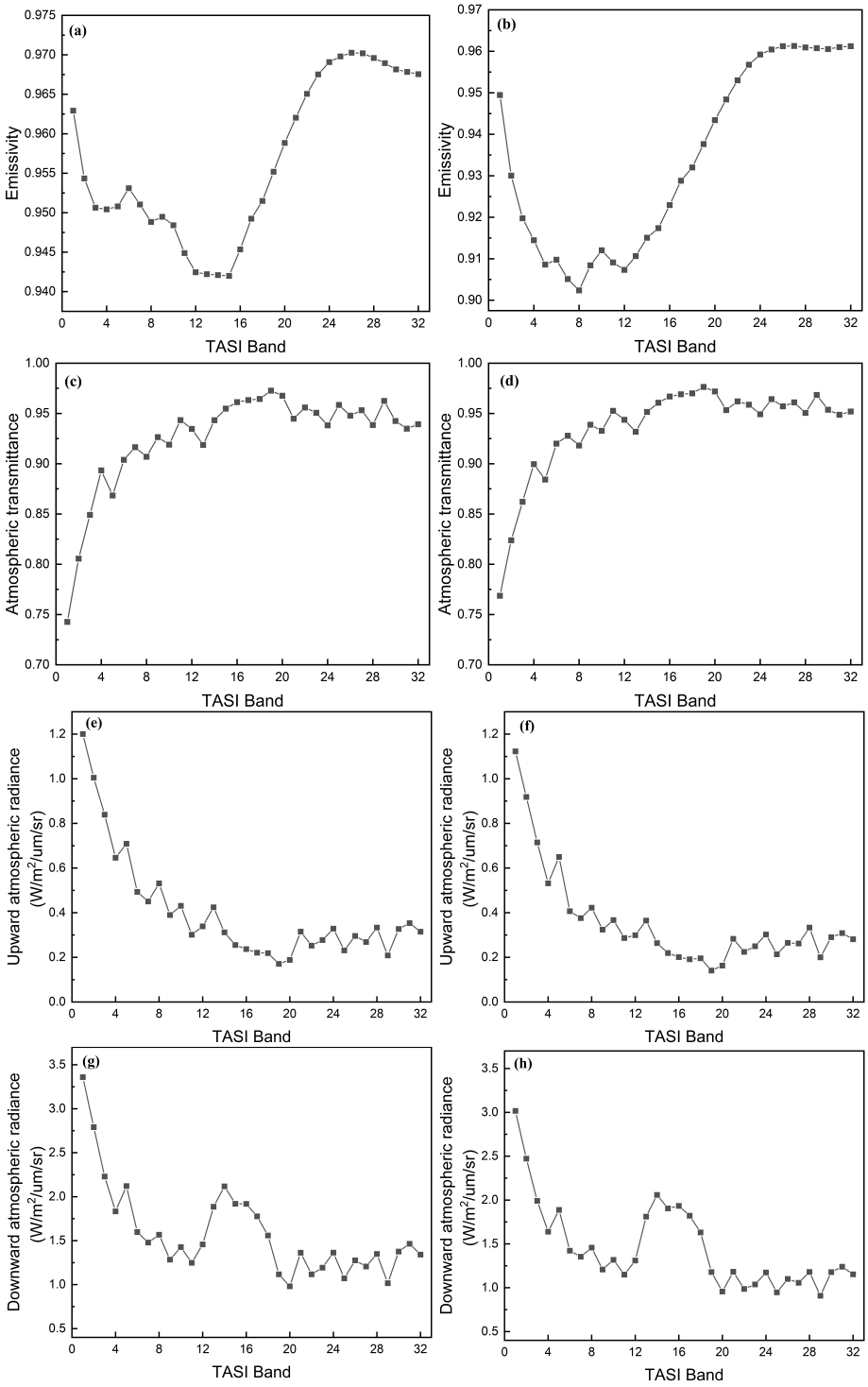


Fig. 11. Retrieval results of each TASI band of the sampling points, emissivity of (a) P1 and (b) P2, atmospheric transmittance of (c) P1 and (d) P2, atmospheric upward radiance of (e) P1 and (f) P2, and atmospheric downward radiance of (g) P1 and (h) P2.

used to obtain ground emissivity with the TASI flight [53], [54]. Because the wavelength ranges between TASI and CE312 were different, the TASI narrowband emissivity was transformed to CE312 bands with the linear empirical function of the following equation to verify the accuracy of the FebLihy algorithm [54]:

$$\varepsilon_i^{\mathrm{CE312}} = \sum_{j=1}^{32} a_{ij}\varepsilon_j^{\mathrm{TASI}} + a_{i0} \tag{21}$$

where $\varepsilon_i^{\mathrm{CE312}}$ $(i = 1, 2, \ldots, 5)$ is the band emissivity of CE312, $\varepsilon_j^{\mathrm{TASI}}$ $(j = 1, 2, \ldots, 32)$ is the band emissivity of TASI, and $a_{ij}$ is the empirical coefficient. The corresponding relationship of band fitting is as follows: $(\varepsilon_3^{\mathrm{TASI}}, \ldots, \varepsilon_5^{\mathrm{TASI}})$ were used to fit $\varepsilon_1^{\mathrm{CE312}}$, $(\varepsilon_5^{\mathrm{TASI}}, \ldots, \varepsilon_8^{\mathrm{TASI}})$ were used to fit $\varepsilon_2^{\mathrm{CE312}}$, $(\varepsilon_9^{\mathrm{TASI}}, \ldots, \varepsilon_{13}^{\mathrm{TASI}})$ were used to fit $\varepsilon_3^{\mathrm{CE312}}$, $(\varepsilon_{21}^{\mathrm{TASI}}, \ldots, \varepsilon_{27}^{\mathrm{TASI}})$ were used to fit $\varepsilon_4^{\mathrm{CE312}}$, and $(\varepsilon_{27}^{\mathrm{TASI}}, \ldots, \varepsilon_{32}^{\mathrm{TASI}})$ were used to fit $\varepsilon_5^{\mathrm{CE312}}$. Thus, the validation was only conducted in the emissivity of CE312 five bands.

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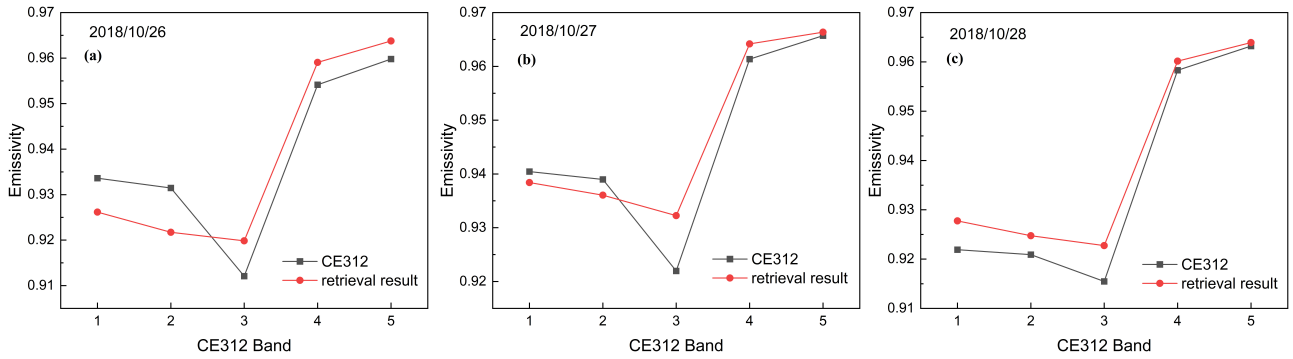


Fig. 12. Emissivity verification between ground measurement data and TASI image retrieval results. (a) October 26, 2018. (b) October 27, 2018. (c) October 28, 2018.

The ground emissivity was considered unchanged in the three-day measurement period because no rainfall or snow occurred during the three adjacent days. The average ground-measured emissivity value of all sampling points on the same day was used to reduce the instrument measurement error. Fig. 12 shows the comparison between CE312 ground measurement and TASI retrieval emissivity. The emissivity retrieval results are very close to the CE312 measurement data. The retrieval results are larger than the measured data in bands 3–5, and the maximum error occurs at CE312 band 3 on October 27, which is approximately 0.0103. The shapes of the emissivity curve in the results are consistent with the ground measurement data. The emissivity of the third band is the smallest, and the fifth band is the largest. The accuracy of the fourth and fifth bands is better than that of the other bands. This finding is reasonable because the thermal radiance of the ground surface (∼300 K) is concentrated in the two bands, with a high signal-to-noise ratio. The wavelength ranges of bands 1–2 of CE312 are located at the edge of the atmospheric window, and atmospheric residual information has a great influence on the accuracy of emissivity retrieval. In summary, land surface emissivity was retrieved accurately with the RMSEs less than 0.01 by using the proposed algorithm.

## V. Discussion

This study focused on the development of a simultaneous retrieval algorithm of atmospheric parameters, LST, and emissivity from aerial TASI hyperspectral thermal infrared images. Accurate results were achieved in the simulated data sets and real TASI images by using the proposed FebLihy algorithm. However, several issues require additional investigations to further improve the retrieval performance.

### A. Possible Improvements of the Retrieval Algorithm

The DNN model used in this study is a simple structured multilayer perceptron. The relationship between each TASI band and emissivity curve shape was not considered in this model. Many complex neural network models, such as the long short-term memory model, will be used to fully exploit the correlation of hyperspectral thermal infrared data for obtaining better results. This study reduced the number of unknowns by fitting a linear empirical function between the different TASI bands for emissivity and atmospheric parameters. However, the fitting residual introduces errors because the emissivity of different land types and the atmospheric parameters of different bands vary greatly. The min–max difference module of the TES algorithm can be used to build a precise emissivity empirical relationship. The initial values of land surface and atmospheric parameters based on the DNN model can be used to derive classical algorithms for estimating land or atmospheric parameters. Inputting initial emissivity into the split-window algorithm to retrieve LST. The initial atmospheric parameters can be used for atmospheric correction in the TES algorithm.

Supplementary data help to improve the FebLihy algorithm. The addition of more land spectral databases and atmospheric profiles to the simulation data set promotes the robustness of the proposed algorithm and makes it suitable for different complicated land surfaces and atmospheric conditions. Land surface classification data can be another important addition. The physical model is expected to further improve the accuracy through targeted optimization of different surface types by analyzing the emissivity patterns.

### B. Further Validation of LST and Atmospheric Parameters

Only a few surface types were validated using CE312 due to the limitations of field experiments. Additional validations will be conducted over different land types, such as cropland, urban, forest, and water. Many field experiments will be conducted in different areas to test the adaptability of the proposed algorithm under different atmospheric conditions. Real atmospheric parameters will be measured concurrently with aerial remote sensing observations to verify the atmospheric results of the FebLihy algorithm. Detailed emissivity measurement data, including many thermal infrared bands collected by other instruments, will be considered in future work to accurately compare the emissivity retrieval results.

## VI. Conclusion

This study proposed the FebLihy algorithm to retrieve LST, emissivity, and atmospheric parameters (transmittance, upward radiance, and downward radiance) simultaneously from the hyperspectral thermal infrared image. The DNN and physical models were combined, and the TASI airborne imager with 32 TIR bands in 8.0–11.5 $\mu$m was used. The FebLihy algorithm first linearized the radiative transfer equation through first-order partial derivative and quadratic fitting of the Planck's law. The number of unknowns was reduced through the linear fitting of the feature bands. A DNN model

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with four hidden layers was used to estimate a group of parameter values without any prior knowledge. The DNN model results were inputted as the initial values into the physical model, and the final retrieval results were achieved after residual minimum optimization.

The FebLihy algorithm was investigated using the simulation data set. The results showed that the retrieval accuracy increases significantly by the physical model compared with the initial values from the DNN model, with the RMSE of LST decreasing from 0.98 to 0.86 K, and the RMSEs of all band emissivity lower than 0.015. For atmospheric parameters, the RMSEs of transmittance decrease from [0.0089, 0.0374] to [0.0082, 0.0331], the RMSEs of upward radiance decrease from [0.0623, 0.2488] to [0.0564, 0.2116] $W/m^2/\mu m/sr$, and the RMSEs of downward radiance decrease from [0.2698, 0.5919] to [0.2528, 0.5070] $W/m^2/\mu m/sr$. The proposed algorithm was applied in the airborne TASI image of Fuyun County in Xinjiang Province and obtained a reliable retrieval image. The retrieval emissivity was validated by ground measurement data using CE312. The results showed that the RMSE is less than 0.01, and the curve shapes of the emissivity retrieval results and the field measurement data are similar.

The proposed FebLihy algorithm has the potential for further improvements in terms of model and data. More sophisticated models and other auxiliary data are expected to reduce errors and improve the accuracy of the algorithm. In the future, we plan to conduct further experiments to validate the results of LST, emissivity, and atmospheric parameters under different atmospheric and land surface conditions. For land surface parameters, more precise LST and emissivity will be measured using hyperspectral TIR instruments instead of the multiband radiometer CE312 used in this article, for instance, 102F Fourier transform infrared (FTIR) spectrometer [55]. For atmospheric parameters, some sounding data synchronized with the TASI observation fight will be collected. Some public data sets, for example, Cloudless Land Atmosphere Radiosounding (CLAR) data [56], will also be taken into account to calculate the atmospheric transmittance, downward, and upward radiance as the validation reference data set.

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